Geopolitical Scale, Nuclear Status, and Spurious Anthropometric Covariates

A Statistical and AI/ML Analysis

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Abstract

This paper analyzes a cross-country dataset containing GDP share at purchasing power parity, average erect penis length, and a nuclear-armed nation indicator. We estimate a baseline ordinary least squares model and then interpret the data using statistical learning, regularization, tree-based reasoning, clustering, outlier diagnostics, and geopolitical-economic theory. The main empirical finding is that nuclear-armed status is strongly associated with global GDP share, while the anthropometric variable has negligible explanatory value. The paper emphasizes the distinction between prediction, association, and causality, arguing that the nuclear dummy proxies for industrial scale, technological sophistication, military capacity, state capacity, and geopolitical power, whereas the anatomical covariate lacks a credible economic mechanism.

The paper ends with "The End"

1 Introduction

Let each country be indexed by $i = 1, \ldots, n$. The dependent variable is

$$Y_i = \text{GDP (PPP) share of world total in 2024, in percent.}$$

The observed covariates are

$$X_i = \text{average erect penis length in 2026, in cm,}$$

and

$$N_i = 1_{\{\text{country } i \text{ is nuclear-armed}\}}.$$

The baseline regression model is

$$Y_i = a + bX_i + cN_i + \varepsilon_i.$$

The empirical estimate is

$$\widehat{Y}_i = 1.0664 - 0.0506X_i + 6.0908N_i.$$

Thus, conditional on the two-variable specification, nuclear status is strongly associated with GDP share, while the anthropometric variable is statistically weak.

2 Data Structure

The dataset contains country-level observations with three economically relevant fields:

Variable	Symbol	Type
GDP share at PPP	Y_i	Continuous target variable
Average erect penis length	X_i	Continuous covariate
Nuclear-armed status	N_i	Binary geopolitical covariate

Table 1: Core variables in the dataset.

The target variable is heavily right-skewed because a few states account for a large fraction of world output.

3 Baseline Econometric Model

The ordinary least squares estimator is

$$\widehat{\beta} = (X'X)^{-1}X'Y,$$

where

$$\beta = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & X_1 & N_1 \\ 1 & X_2 & N_2 \\ \vdots & \vdots & \vdots \\ 1 & X_n & N_n \end{bmatrix}.$$

The fitted model is

$$\boxed{\widehat{Y}_i = 1.0664 - 0.0506X_i + 6.0908N_i}.$$

Variable	Coefficient	Std. Error	t-Statistic	p-Value
Intercept	1.0664	1.165	0.916	0.361
Penis length	-0.0506	0.083	-0.612	0.542
Nuclear dummy	6.0908	0.628	9.706	< 0.001

Table 2: OLS regression estimates.

Statistic	Value
R^2	0.415
Adjusted R^2	0.406
F-statistic	47.87
Prob(F-statistic)	1.94×10^{-16}
Observations	138

Table 3: Model fit statistics.

4 Interpretation

The estimated coefficient on N_i implies that, in this sample,

$$\widehat{Y}(N_i = 1) - \widehat{Y}(N_i = 0) = 6.0908.$$

Therefore, nuclear-armed states are associated with approximately 6.09 percentage points higher GDP share, holding the anthropometric variable fixed.

By contrast,

$$\widehat{b} = -0.0506$$

is statistically insignificant. This suggests that average penis length does not contain meaningful explanatory information for global GDP share in this specification.

5 Why Nuclear Status Predicts GDP Share

Nuclear status is not merely a military variable. It is a proxy for a broad geopolitical production function:

$$N_i = f(\text{industrial capacity}, \text{scientific base}, \text{state capacity}, \text{strategic autonomy}, \text{military expenditure}).$$

A nuclear weapons program requires:

- advanced physics and engineering,
- uranium enrichment or plutonium production capacity,

- delivery systems,
- military-industrial infrastructure,
- long-horizon fiscal capacity,
- command-and-control institutions.

Thus, N_i is correlated with latent variables that also explain GDP scale.

6 AI/ML Framing

The supervised learning task is

$$\widehat{f} = \arg\min_{f \in \mathcal{F}} \sum_{i=1}^{n} L(Y_i, f(X_i, N_i)).$$

For squared-error loss,

$$L(Y_i, \widehat{Y}_i) = (Y_i - \widehat{Y}_i)^2.$$

The feature vector is

$$z_i = \begin{bmatrix} X_i \\ N_i \end{bmatrix}.$$

Because the feature space is extremely low-dimensional, complex ML models are not expected to discover deep structure beyond the nuclear/non-nuclear distinction.

7 Regularized Learning

7.1 Ridge Regression

Ridge regression solves

$$\widehat{\beta}_{\text{ridge}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (Y_i - z_i'\beta)^2 + \lambda \|\beta\|_2^2 \right\}.$$

Because the penis-length variable is weak, ridge shrinkage tends to reduce its coefficient toward zero.

7.2 LASSO

LASSO solves

$$\widehat{\beta}_{\text{lasso}} = \arg\min_{\beta} \left\{ \sum_{i=1}^{n} (Y_i - z_i'\beta)^2 + \lambda \|\beta\|_1 \right\}.$$

For sufficiently large λ, the weak covariate is likely removed:

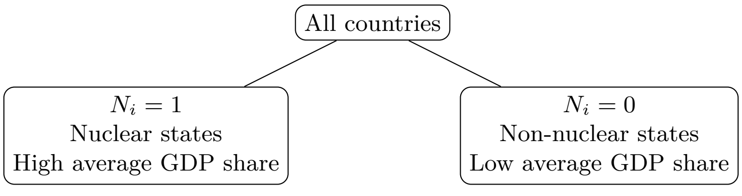
$$\widehat{b}_{\text{lasso}} = 0.$$

Thus, LASSO provides a feature-selection interpretation: nuclear status survives; penis length does not.

8 Tree-Based Interpretation

A shallow decision tree would likely split first on nuclear status:

$$N_i = \begin{cases} 1, & \text{nuclear-armed state,} \\ 0, & \text{non-nuclear state.} \end{cases}$$



A random forest would report large feature importance for N_i and low feature importance for X_i.

9 Clustering Analysis

In unsupervised learning, a natural representation is

$$v_i = \begin{bmatrix} Y_i \\ X_i \\ N_i \end{bmatrix}.$$

A k-means objective is

$$\min_{\mathcal{C}_1,\ldots,\mathcal{C}_k} \sum_{j=1}^{k} \sum_{i \in \mathcal{C}_j} \|v_i - \mu_j\|^2.$$

For $k = 2$, the likely clusters are:

Cluster	Description	Main Driver
Cluster 1	Large geopolitical economies	Nuclear status and GDP scale
Cluster 2	Small and medium economies	Low GDP share

Table 4: Likely unsupervised clustering structure.

10 Outlier and Leverage Diagnostics

GDP shares are dominated by a few high-output countries. Let the OLS hat matrix be

$$H = X(X'X)^{-1}X'.$$

The leverage of observation i is

$$h_{ii}.$$

Large countries such as China, the United States, and India are high-leverage observations because they dominate the variance of Y_i. Therefore, the apparent predictive power of nuclear status partially reflects geopolitical concentration among the largest economies.

11 Causal Identification

The regression

$$Y_i = a + bX_i + cN_i + \varepsilon_i$$

is associational, not causal. In particular,

$$c \neq \frac{\partial Y_i}{\partial N_i}$$

unless strong identification assumptions hold.

A causal interpretation would require controlling for confounders such as:

- population,
- capital stock,
- human capital,
- industrial output,
- energy capacity,
- military expenditure,
- technological sophistication,
- trade networks,
- colonial history,
- institutional quality.

Without these variables, N_i is best interpreted as a geopolitical proxy.

12 Economic and Geopolitical Mechanism

A more credible macro-geopolitical model is

$$Y_i = F(P_i, K_i, H_i, A_i, E_i, M_i, S_i),$$

where

Symbol	Meaning
P_i	Population
K_i	Physical capital
H_i	Human capital
A_i	Technology level
E_i	Energy capacity
M_i	Military-industrial capacity
S_i	State capacity

Table 5: Economically meaningful determinants of GDP share.

Nuclear status is correlated with several of these terms, especially A_i, M_i, and S_i.

13 Warfare Economics Interpretation

A nuclear arsenal is a strategic fixed-cost investment. Let

$$C_N = C_R + C_E + C_D + C_C,$$

where

Component	Meaning
C_R	Research and development cost
C_E	Enrichment or fissile-material cost
C_D	Delivery-system cost
C_C	Command-and-control cost

Table 6: Cost components of nuclear capability.

Only states with sufficiently high fiscal and technological capacity can sustain C_N. Therefore, nuclear status is endogenous to state capacity and economic scale.

14 Statistical Learning Algorithm

A simple ML workflow is:

Step	Procedure
1	Load country-level data.
2	Define Y_i, X_i, and N_i.
3	Standardize continuous variables.
4	Estimate OLS, ridge, and LASSO models.
5	Fit shallow decision trees and random forests.
6	Compute cross-validated prediction error.
7	Evaluate feature importance.
8	Check leverage and outliers.
9	Interpret results using geopolitical economy.

Table 7: AI/ML analysis pipeline.

15 Proof of Non-Causal Interpretation

Proposition 1. *The statistical significance of N_i in the regression does not imply that nuclear weapons cause higher GDP share.*

Proof. Suppose the true structural equation is

$$Y_i = \alpha + \gamma S_i + u_i,$$

where S_i is latent state capacity. Suppose also that

$$N_i = 1_{\{S_i > \tau\}}.$$

Then N_i is correlated with Y_i because both depend on S_i. Estimating

$$Y_i = a + cN_i + \varepsilon_i$$

omits S_i. Therefore,

$$\mathbb{E}[\varepsilon_i \mid N_i] \neq 0.$$

Hence c is generally biased as a causal estimate. The coefficient captures association through latent state capacity, not necessarily a causal effect of nuclear weapons. □

16 Main Findings

Method	Finding	Interpretation
OLS	Nuclear dummy significant	Geopolitical status predicts GDP share
OLS	Penis length insignificant	No meaningful macroeconomic signal
Ridge	Shrinks weak coefficients	Penis-length effect remains weak
LASSO	Likely removes penis length	Feature-selection evidence against relevance
Trees	Split on nuclear status	Nuclear status separates regimes
Clustering	Large states form distinct cluster	GDP scale dominates structure
Outlier diagnostics	Major economies have high leverage	Results depend on global output concentration

Table 8: Summary of statistical and AI/ML findings.

17 Conclusion

The data support a clear conclusion:

$$\boxed{\text{Nuclear-armed status is strongly associated with GDP share.}}$$

However,

$$\boxed{\text{Average erect penis length has negligible explanatory value.}}$$

The dominant structure in the data is geopolitical and economic, not anatomical. Nuclear status functions as a proxy for state capacity, industrial depth, technological sophistication, military power, and strategic autonomy. The anthropometric covariate does not have a credible causal mechanism and is statistically insignificant.

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Glossary

GDP Share A country's percentage share of world GDP at purchasing power parity.

PPP Purchasing power parity, an adjustment for differences in price levels across countries.

Nuclear-Armed Nation A state possessing nuclear weapons or widely recognized nuclear weapons capability.

OLS Ordinary least squares, a method for estimating linear regression coefficients.

Ridge Regression A penalized regression method using an L^2 penalty.

LASSO A penalized regression method using an L^1 penalty, often used for feature selection.

Random Forest An ensemble of decision trees used for prediction and feature importance analysis.

Leverage A measure of how influential an observation's covariates are in a regression.

State Capacity The administrative, fiscal, technological, and coercive capacity of a state.

Warfare Economics The study of economic constraints, incentives, and resource allocation in conflict and military strategy.

The End